

Per-dataset layer-wise probing: ID vs OOD (redraw + re-check)

Following the feedback that the earlier figures collapsed each dataset into a single number, here are **per-dataset**, **per-layer** curves so the ID→OOD change is visible at every layer — plus a per-dataset re-check of whether the tunnel shape appears in each.

Figures - `fig_grid_id_tunnel` — ID probe accuracy by layer, one panel per dataset → *the tunnel shape*. - `fig_grid_idood_gauss` / `_timewarp` / `_drift` — ID vs each OOD shift, per dataset. - `fig_grid_idood_all` — ID + all three shifts overlaid (overview). - `perdataset_summary.json` — all numbers + raw per-layer arrays.

8 datasets, ordered strong → weak → saturated reference. Probe = linear logistic regression per layer on frozen Chronos-2 features (StandardScaler on train only); 95% bootstrap CIs; “late-drop” = pre-declared middle band (L3–8) minus last layer (L11). Epilepsy & Cricket are saturated (flat ID curves at ceiling) — shown as reference only, excluded from the claims.

Finding 1 — the tunnel shape holds, per dataset

The ID curves show the rise → plateau → late-drop shape in every headroom dataset, with the peak (argmax) sitting in the **middle band (L3–L7)**, **not at the last layer**. The last-layer deficit (mean L3–8 – L11) is positive with its 95% CI excluding 0 in **5 of 6** non-saturated datasets:

dataset	domain	chance	late-drop (ID), 95% CI	sig?	argmax
UWaveGestureLibrary	gesture	.125	+0.085 [.051, .122]	yes	L4
EthanolConcentration	chem. spectra	.250	+0.070 [.013, .122]	yes	L3
SelfRegulationSCP1	EEG	.500	+0.063 [.028, .098]	yes	L4
Handwriting	handwriting	.038	+0.050 [.026, .075]	yes	L6
LSST	light-curves	.071	+0.015 [.000, .029]	yes (borderline)	L7
SelfRegulationSCP2	EEG	.500	−0.018 [−.091, .058]	no	—

- **SCP2** is the lone null — its probe sits near chance at every layer, so it’s *underpowered*, not a counterexample.
- **LSST** is borderline (lower CI edge at 0.000) — a real but very small effect, resolvable only because its test set is large (2466).

→ Across EEG, handwriting, gesture, spectra and light-curves, the middle layers carry more linearly-decodable class information than the final layer.

Finding 2 — OOD damage is shift-dependent and not last-layer-concentrated (amplification null)

The redrawn grids reveal that the OOD curves do **not** behave uniformly — the shape depends on the shift, and the damage is generally *not* concentrated at the last layer (which is exactly what Part 2 predicted):

Per-dataset ID layer-wise probe accuracy (tunnel shape?) - 95% bootstrap CI

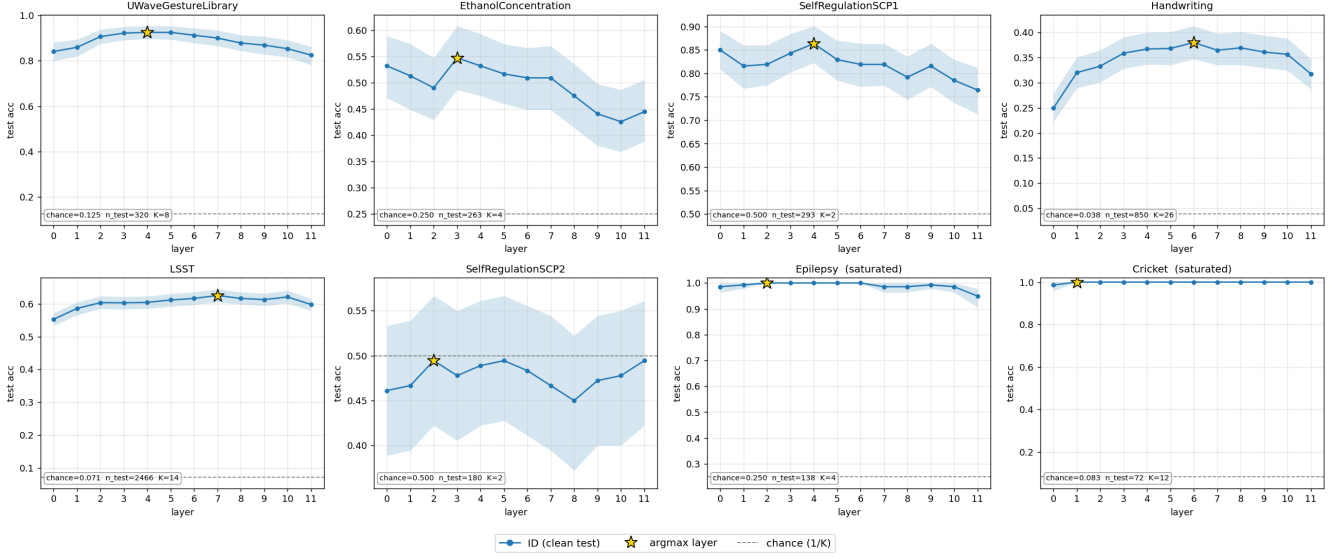


Figure 1: ID layer-wise probe accuracy per dataset; gold star = argmax layer

- **Timewarp** (`fig_grid_idood_timewarp`): the OOD curve sits roughly **parallel** below ID in every headroom dataset — it degrades all layers about equally → no amplification.
- **Gaussian noise** (`fig_grid_idood_gauss`): where resolvable, the damage concentrates at the **early** layers, not the late ones — UWave and SCP1 crater at L0–L1 and recover by the middle. That is the *opposite* of the tunnel prediction, which is why several gauss cells come out **negative** (UWave/gauss -0.082 , sig). The one exception is SCP1, whose last layer also collapses (L11 → 0.63), giving the lone positive gauss cell ($+0.082$). On the hardest tasks (Ethanol, Handwriting) gauss floors the probe near chance at all layers.
- **Drift** (`fig_grid_idood_drift`): mostly mild / overlapping with ID, except Handwriting, where OOD peels away more at the late layers ($+0.040$, the one pro-tunnel drift cell), and Ethanol, which degrades erratically (-0.086 , sig negative).

Tallying the 18 non-saturated amplification cells (mid-vs-last gap, $\text{OOD} - \text{ID}$): **3 positive-significant** (SCP1/gauss, UWave/timewarp, Handwriting/drift), **2 negative-significant** (UWave/gauss, Ethanol/drift), and 13 indistinguishable from 0 — scattered across different datasets and shifts, with as many hits in the *wrong* direction as the right one. That is the signature of no real effect (multiple-comparison noise).

→ The last-layer-preferential degradation Part 2 predicted does not appear. If anything, the most input-like shift (Gaussian) hits the **early** layers hardest — consistent with early layers being closest to the raw input — which is a *different* phenomenon from the tunnel effect.

Bottom line

- **Part 1 (tunnel shape: middle > last) — robust, per dataset.** 5/6 non-saturated datasets, middle argmax (L3–L7), across five distinct domains.
- **Part 2 (gap widens under shift) — null, per dataset and per shift.** The redrawn grids show OOD damage is *not* concentrated at the last layer; its location is shift-dependent, and for Gaussian noise it lands on the *early* layers. The few significant amplification cells are scattered and balanced by equally-significant cells in the opposite direction.
- **Saturated datasets (Epilepsy, Cricket)** are reference only — flat ID curves at ceiling; Cricket’s large negative gauss amplification (-0.236) is just a saturated probe collapsing unevenly, not interpretable.

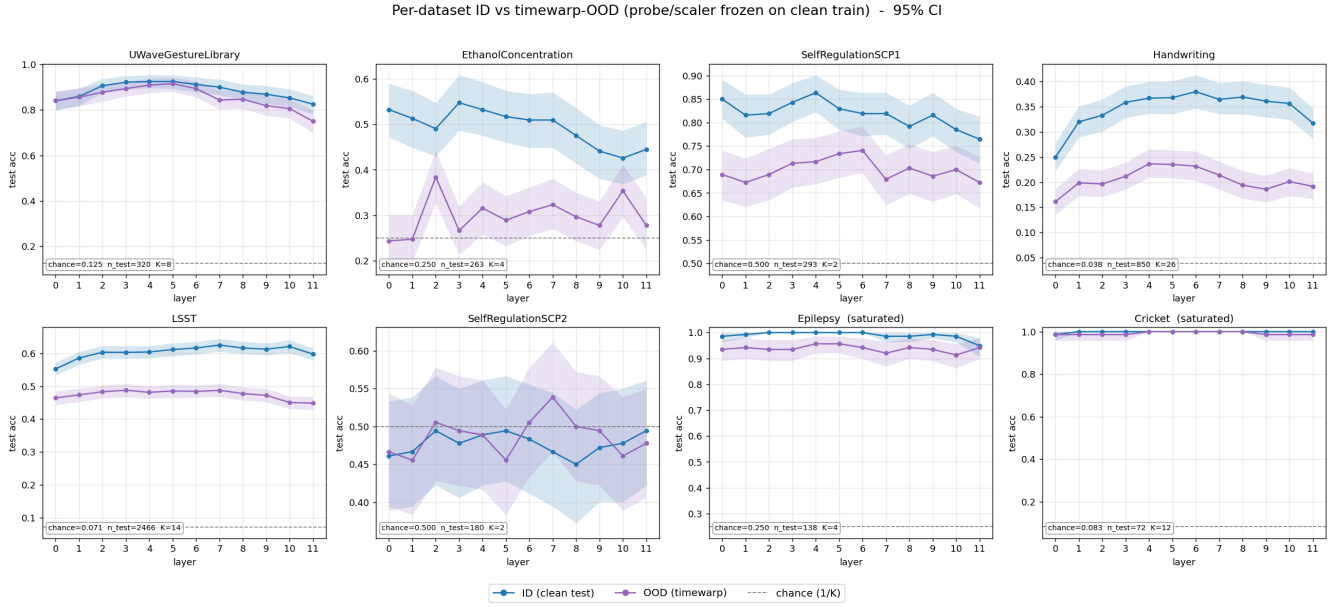


Figure 2: ID vs timewarp-OOD per dataset — the OOD curve sits roughly parallel below ID (uniform degradation, no amplification).

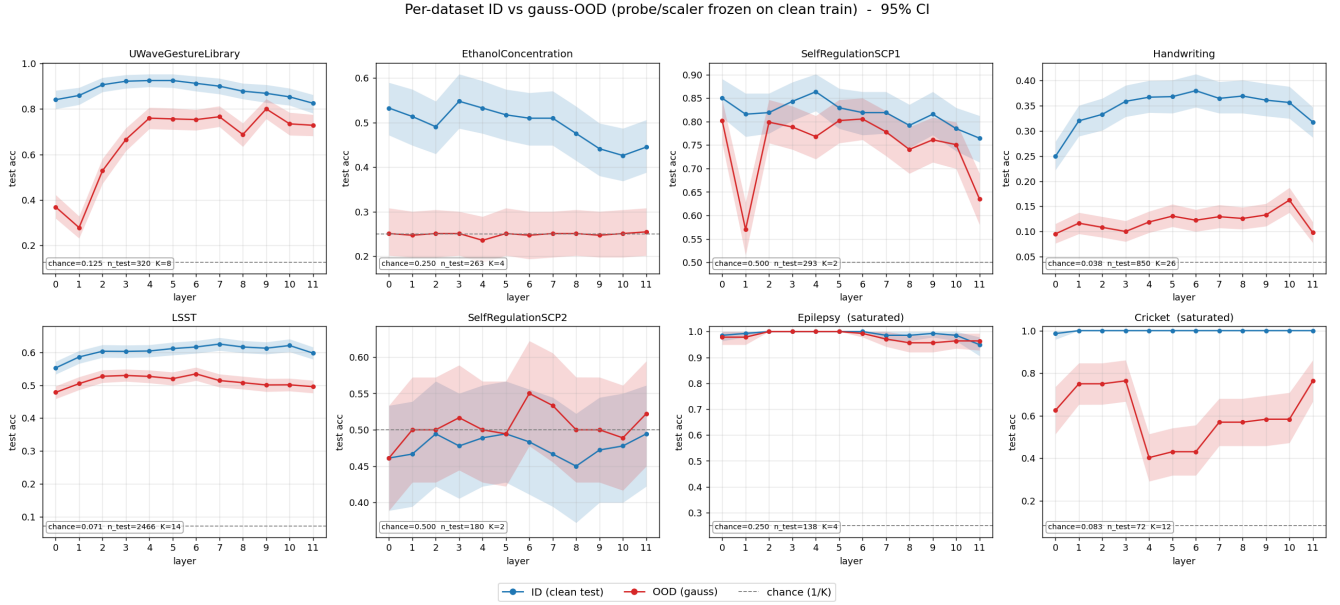


Figure 3: ID vs Gaussian-noise OOD per dataset — where resolvable, damage concentrates at the *early* layers (L0-L1), not the last.

Per-dataset ID vs drift-ODD (probe/scaler frozen on clean train) - 95% CI

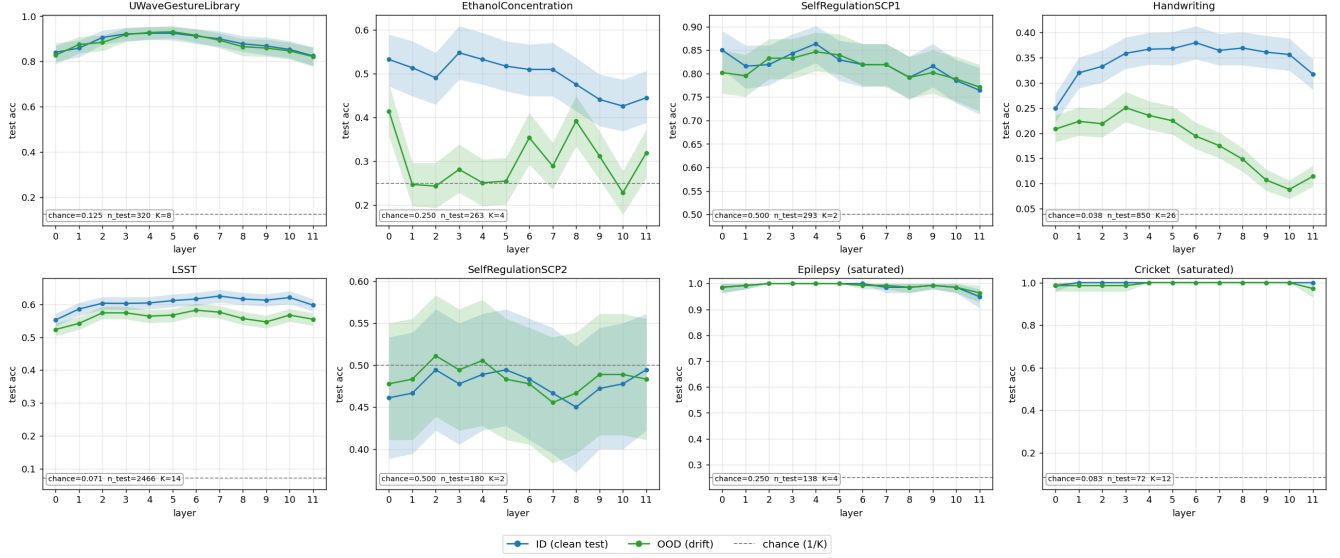


Figure 4: ID vs drift-ODD per dataset — mostly mild / overlapping with ID, except Handwriting (late-layer peel-away) and Ethanol (erratic).

Per-dataset ID + all OOD shifts (overview, no CI bands)

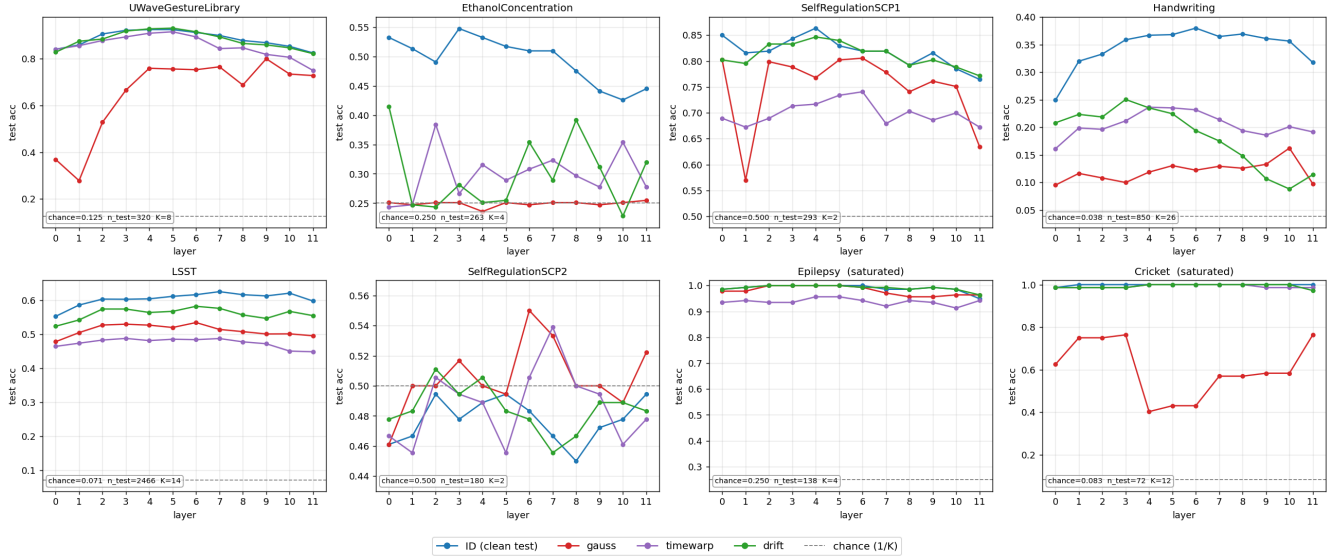


Figure 5: Overview: ID + all three shifts overlaid per dataset (no CI bands, for readability).

Open question (unchanged): the per-dataset tunnel shape could be the tunnel effect, or the model being out-of-distribution on these domains. The clean disambiguation is matched probes on an *in-distribution* vs *out-of-distribution* task — the late-layer deficit should grow with distance from pretraining if it's truly the tunnel effect. The early-layer fragility under Gaussian noise is a separate, input-coupling effect worth a short look on its own.

Numbers and raw per-layer arrays in `perdataset_summary.json`.